INVENTION DISCLOSURE FORM

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1. **Particulars of Inventors**

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2. **Provide a brief descriptive title of the invention:**
   Deep Learning-Based Smart Ration Distribution & Fraud Detection System Using Hybrid Adaptive Identity Verification and Behavioral Analysis

3. **In 100 words or less, please provide an abstract or summary of the invention:**
   This invention proposes an intelligent ration distribution system using deep learning to ensure secure and transparent delivery of subsidized goods. The system integrates multi-modal biometric verification, behavioral pattern analysis, and anomaly-based fraud detection. A hybrid decision engine dynamically evaluates trust scores before approving transactions. The system reduces identity fraud, prevents misuse of ration resources, and ensures real-time monitoring with secure data storage. This approach significantly improves efficiency, reliability, and accountability in public distribution systems.

4. **Detail description of the invention:(Answer to all below are required in detail)**

   a. **Problem the invention is solving**

   The present invention addresses major issues in traditional ration distribution systems.There is a high risk of identity fraud due to the use of fake or duplicate rationcards.Manual record maintenance leads to errors, manipulation, and corruption.Existing systems lack real-time monitoring and intelligent fraud detection mechanisms.Unauthorized users often exploit loopholes to access subsidized resources.Hence, there is a need for a secure, automated, and intelligent ration distribution system.

   b. **General Utility/application of the invention**

   The invention can be used in government public distribution systems for fair ration allocation.It is applicable in welfare schemes that require beneficiary verification.The system can be deployed in smart city infrastructures for automated resource distribution.It can also be extended to subsidy management systems in various sectors.The invention improves transparency, accountability, and efficiency in resource allocation.

   c. **Advantages of the invention disclosing about the increased efficiency/efficacy**

   The invention provides highly accurate identity verification using deep learning techniques.It reduces fraud by detecting abnormal behavior patterns and suspicious activities.The system enables real-time monitoring and automated decision-making.It minimizes human intervention, thereby reducing corruption and manual errors.Secure data storage ensures protection against tampering and unauthorized access.The system improves efficiency, speed, and reliability of ration distribution.

   d. **Best way of using the invention as well as possible variants**

   The system works best when integrated with biometric authentication such as face recognition.It can be enhanced by adding fingerprint or iris recognition as additional verification layers.The system can be deployed on cloud infrastructure for scalability and accessibility.A mobile application interface can be developed for remote monitoring and administration.The invention can be customized based on regional requirements and government policies.

   e. **Working of invention along with Drawing, schematics and flow diagrams if required with complete explanations**

   The working of the invention begins when the user visits the ration distribution center.The system captures the user’s facial image using a camera.The captured data is preprocessed to remove noise and normalize input.

The identity verification module (HMIV-Net) analyzes facial features and verifies the user.The system performs liveness detection to ensure that the input is from a real person.A confidence score is generated to indicate the authenticity of the user.
The behavioral analysis module (ARUPLA) evaluates the user's past transaction history.It identifies patterns such as frequency of visits and quantity of ration consumed.The system detects deviations from normal behavior using sequence learning models.
The fraud detection module (HAD-RSM) processes the input data for anomaly detection.It combines deep learning techniques and rule-based methods to identify suspicious activities. A fraud risk score is generated based on the detected anomalies.
The decision engine (DTBRAA) combines identity confidence, behavior score, and fraud risk score.It calculates a trust score using a weighted formula.
Based on the trust score, the system decides whether to approve, alert, or reject the request.
Finally, the transaction is securely stored using encrypted protocols in the database.All actions are logged for auditing and monitoring purposes.

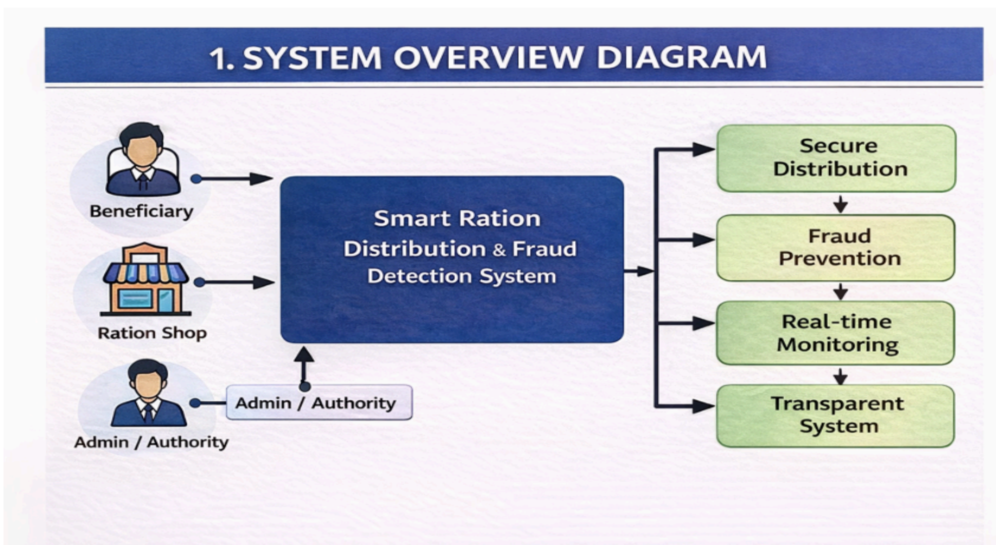


Figure 1: Diagram of System overview

The system overview diagram shows the complete workflow of the smart ration distribution system from user input to final ration allocation.
It captures user data, performs identity verification using deep learning, and analyzes behavior to detect abnormal patterns.
A fraud detection module evaluates risk, and a decision engine determines whether to approve or reject the transaction.
Finally, the transaction is securely stored, and ration is distributed only to verified and eligible users.

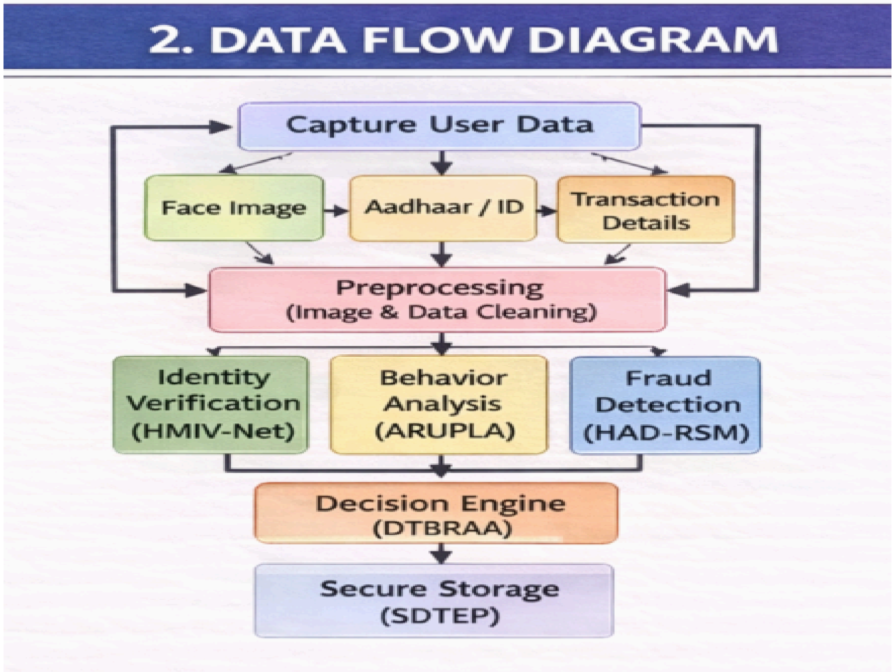


Figure 2: Diagram of Data flow

The data flow diagram illustrates how user data moves through the smart ration system from input to final storage.
It shows the capture of face image, ID details, and transaction data, which are preprocessed before being sent to AI modules.
The data is then processed by identity verification, behavior analysis, and fraud detection modules.
Finally, the decision engine produces an output, and the transaction is securely stored in the database.

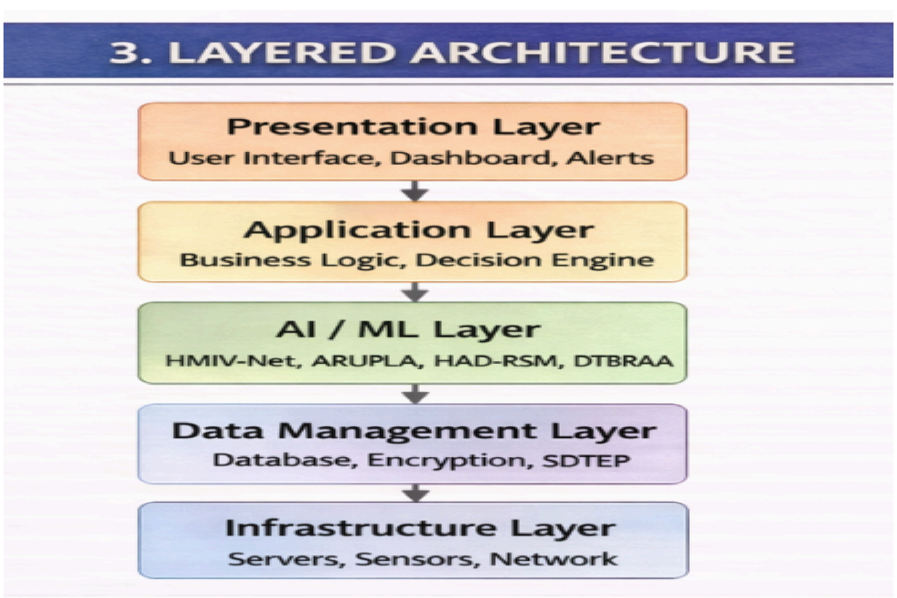


Figure 3: Diagram of Layered Architecture System

The layered architecture diagram presents the system as a stack of functional layers, each responsible for a specific role.
The presentation layer handles user interfaces, while the application layer manages business logic and decision-making.
The AI/ML layer performs identity verification, behavior analysis, and fraud detection.
The data and infrastructure layers ensure secure storage, processing, and system deployment.

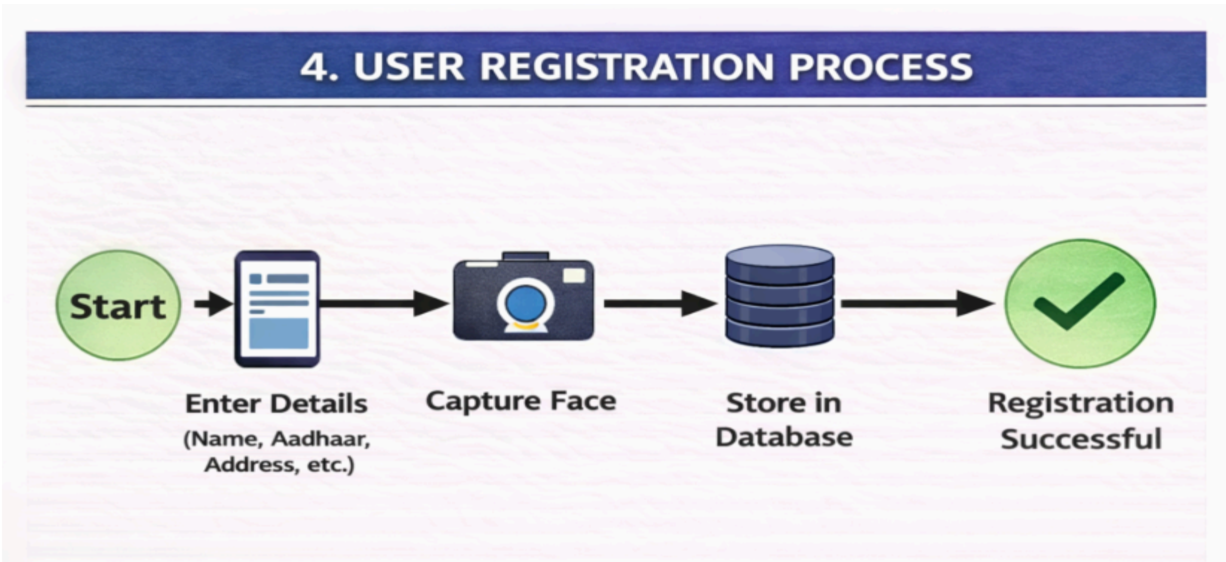


Figure 4: Diagram of User registration process flowchart

The user registration process flowchart shows the steps required to enroll a new beneficiary into the system.
It begins with entering personal details and capturing the user's facial image for biometric identification.
The system then validates the provided information and checks for duplicates.
Once verified, the user data is securely stored in the database and registration is completed successfully.

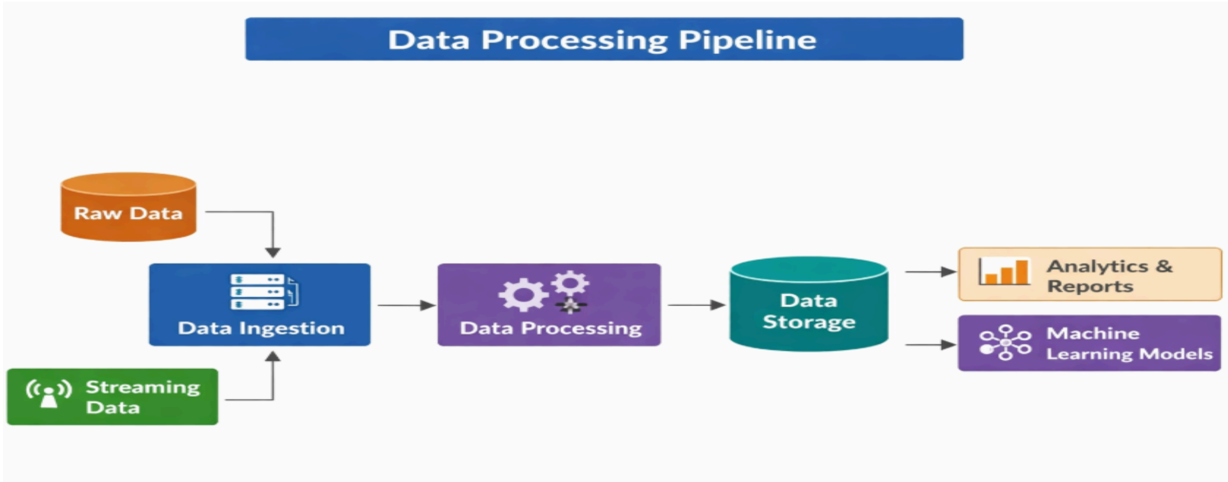


Figure 5: Diagram of data processing pipeline

The data processing pipeline diagram shows how raw input data is transformed into usable information within the system.
It begins with data collection from user inputs and sensors, followed by preprocessing such as cleaning and normalization.
The processed data is then passed through analytical and AI modules for feature extraction and decision-making.
Finally, the results are stored securely and used for further actions like ration allocation or reporting.

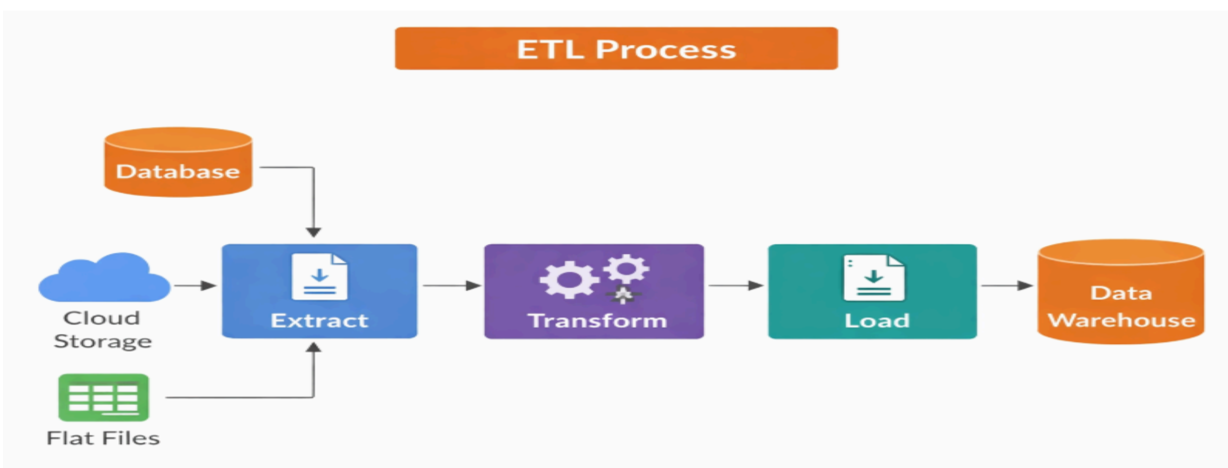


Figure 6: Diagram of ETL process

The ETL process diagram illustrates how data is handled in three stages: Extract, Transform, and Load.
In the extract stage, data is collected from sources such as databases, user inputs, or external systems.
During the transform stage, the data is cleaned, formatted, and processed for analysis.
Finally, in the load stage, the processed data is stored in a database or data warehouse for further use.

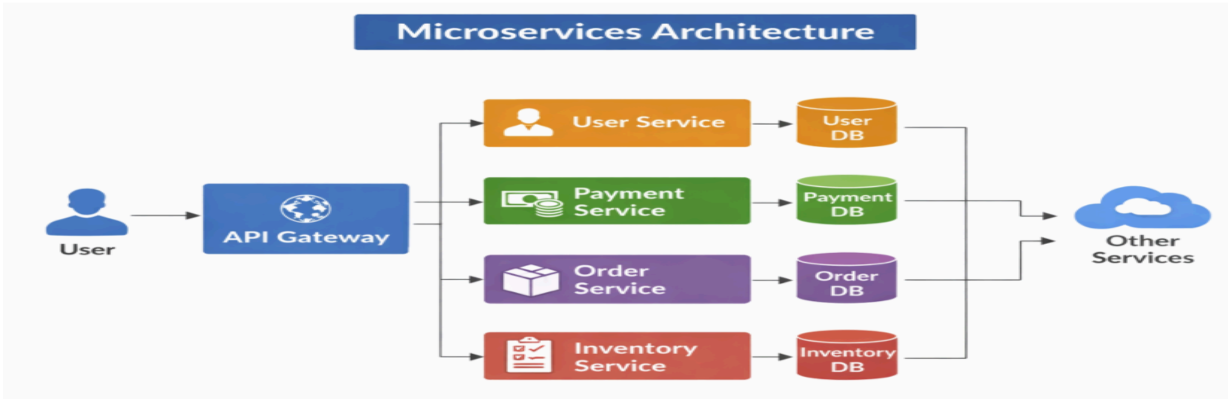


Figure 7: Diagram of Microservices architecture

The microservices architecture diagram shows the system divided into independent, modular services that communicate through APIs.
Each service handles a specific function such as identity verification, behavior analysis, fraud detection, or data management.
An API gateway manages requests and routes them to the appropriate services.
This architecture improves scalability, flexibility, and fault isolation in the smart ration system.

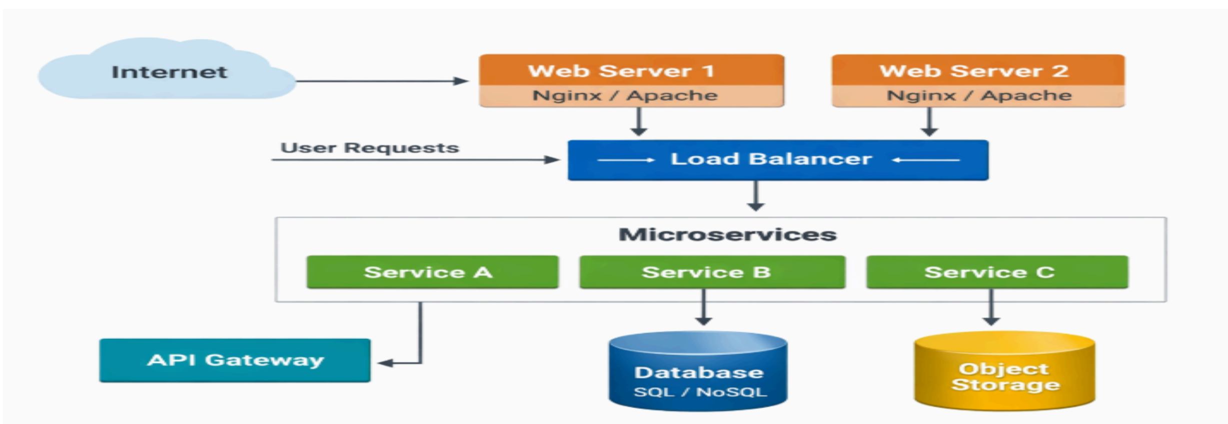


Figure 8: Diagram of Microservices cloud architecture

The microservices cloud architecture diagram shows how independent services are deployed and managed in a cloud environment.
Each microservice runs in containers or virtual instances and communicates through APIs or service mesh.
Cloud components like load balancers, auto-scaling, and storage ensure high availability and scalability.
This setup enables efficient resource utilization, resilience, and seamless system expansion.

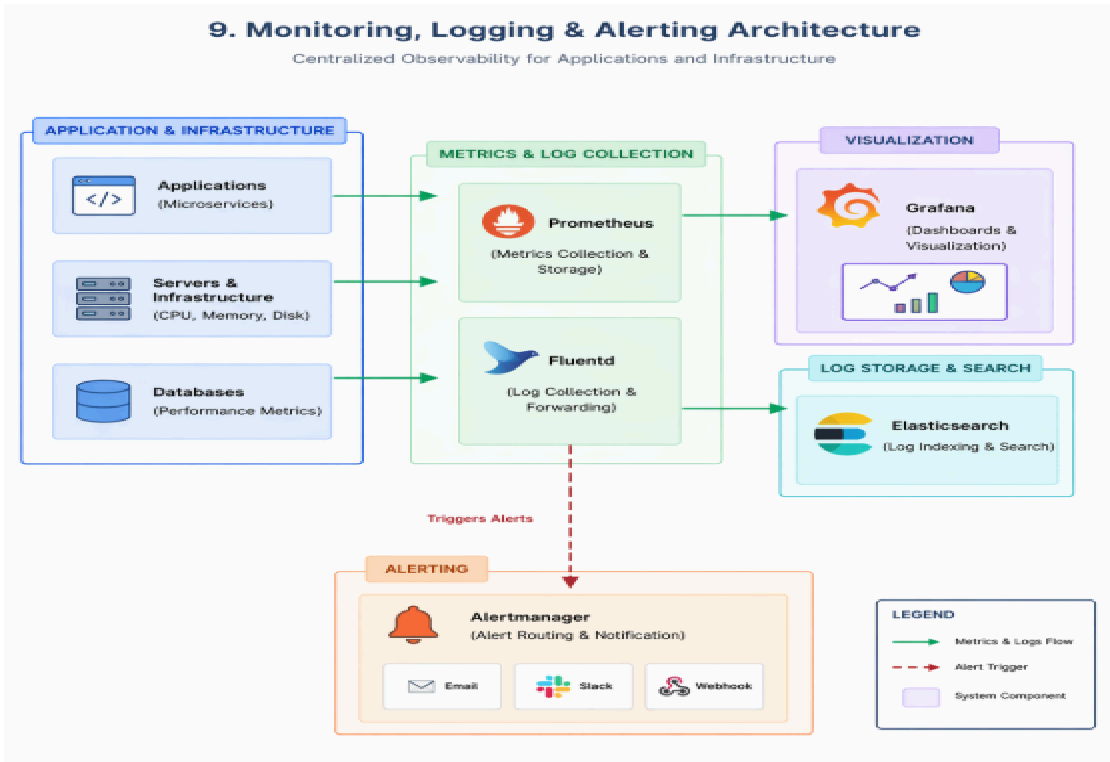


Figure 9: Diagram of Monitoring,Logging & Alerting Architecture

The Monitoring, Logging, and Alerting Architecture diagram represents the framework used to track, analyze, and manage the system's performance and operational activities.
It begins with data collection from various components such as identity verification, behavior analysis, fraud detection modules, and database systems.
These components generate metrics (CPU usage, response time, transaction count) and logs (user actions, system events, errors).
The collected data is forwarded to centralized monitoring and logging systems for aggregation and storage.
The monitoring system continuously evaluates metrics to assess system health and performance.
Logging systems store detailed records that help in debugging and auditing system activities.
Visualization tools such as dashboards present this information in an easy-to-understand graphical format.
Administrators can use these dashboards to monitor real-time system behavior and identify trends.
The alerting mechanism plays a critical role by detecting anomalies or threshold breaches.
Alerts are triggered automatically when unusual patterns or system failures are observed.
Notifications are sent to administrators via email, SMS, or system alerts for immediate action.
This architecture ensures proactive issue detection, minimizes downtime, and improves system reliability.
It also enhances security by identifying suspicious activities and supports efficient maintenance and troubleshooting.

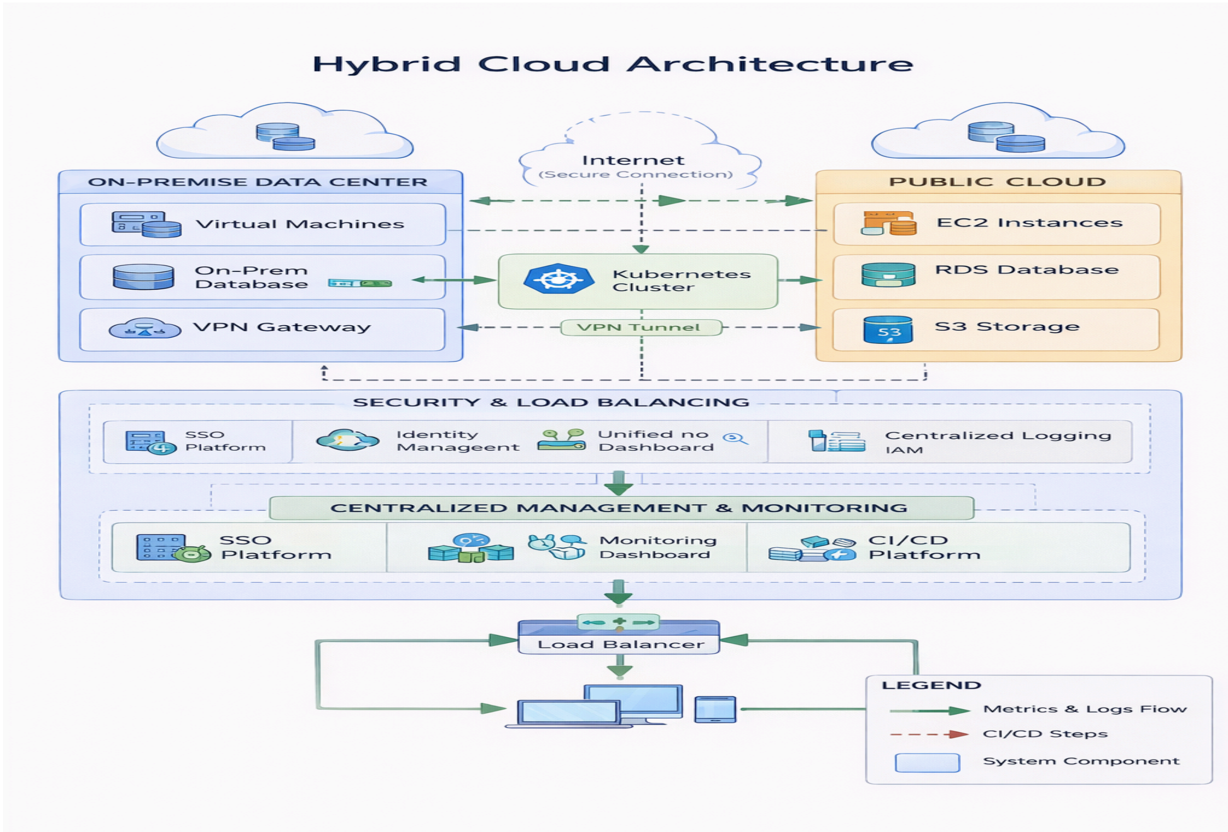


Figure 10: Diagram of Hybrid Cloud Architecture

The Hybrid Cloud Architecture diagram illustrates how the system combines on-premise infrastructure with public and private cloud environments to deliver a flexible and scalable solution.
It shows that critical components, such as sensitive user data and authentication services, can be maintained within the on-premise data center for enhanced security and control.
At the same time, computationally intensive modules like deep learning models and analytics are deployed on the cloud to leverage scalable resources.
The diagram highlights secure communication between on-premise systems and cloud services through technologies such as VPNs or secure APIs.
Data is synchronized between environments to ensure consistency and real-time availability across the system.
Load balancers distribute traffic efficiently between local and cloud servers to optimize performance.
Cloud platforms provide services such as storage, compute instances, and container orchestration for running microservices.
Auto-scaling mechanisms dynamically adjust resources based on demand, ensuring high availability during peak usage.
The architecture also incorporates security layers, including firewalls, identity management, and encryption protocols.
Monitoring and management tools oversee both on-premise and cloud components from a unified interface.

This hybrid approach enables organizations to balance cost, performance, and security requirements effectively.
It allows gradual migration to the cloud while maintaining legacy systems.
Overall, the hybrid cloud architecture ensures reliability, scalability, and efficient resource utilization for the smart ration distribution system.

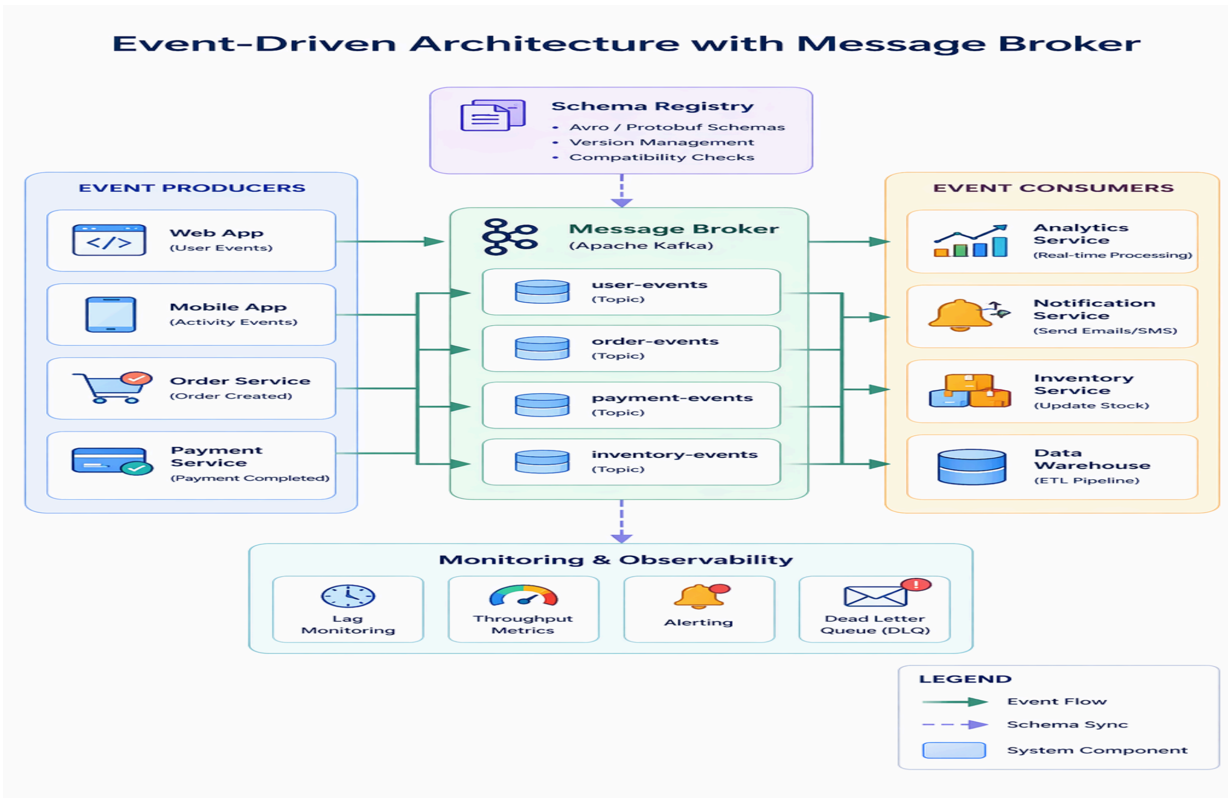


Figure 11: Diagram of Event-Driven Architecture with Message Broker

The event-driven architecture diagram shows how system components communicate asynchronously using a message broker.
Producers generate events (such as user transactions) and publish them to the broker without waiting for a response.
The message broker distributes these events to multiple consumers that process tasks independently.
This approach improves scalability, flexibility, and real-time processing in the smart ration system.

5. **Have you conducted Primary Patent Search? Yes / No (if yes, attach the patent search report)**
No

6. **Existing state-of-the-art and prior arts: (Brief background of the existing knowledge/product/process in the market)**
Existing systems primarily rely on manual verification or basic biometric authentication.
Aadhaar-based authentication systems are commonly used but lack behavioral analysis.
Some systems use OTP-based verification, which can be bypassed or misused.
Current solutions do not integrate deep learning models for fraud detection.
Most systems lack real-time monitoring and adaptive decision-making capabilities.

7. **List out the known ways about how others have tried to solve the same or similar problems? Indicate the disadvantages of these approaches. In addition, please identify any prior art documentation or other material that explains or provides examples of such prior art efforts.**

S. No.	Existing state of art	Drawbacks in existing state of art	Overcome (how your invention is overcoming the drawback)
1	Manual System	Fraud & errors	Automated AI system
2	OTP System	Can be bypassed	Biometric + AI
3	Biometric Only	No behavior analysis	Hybrid AI system

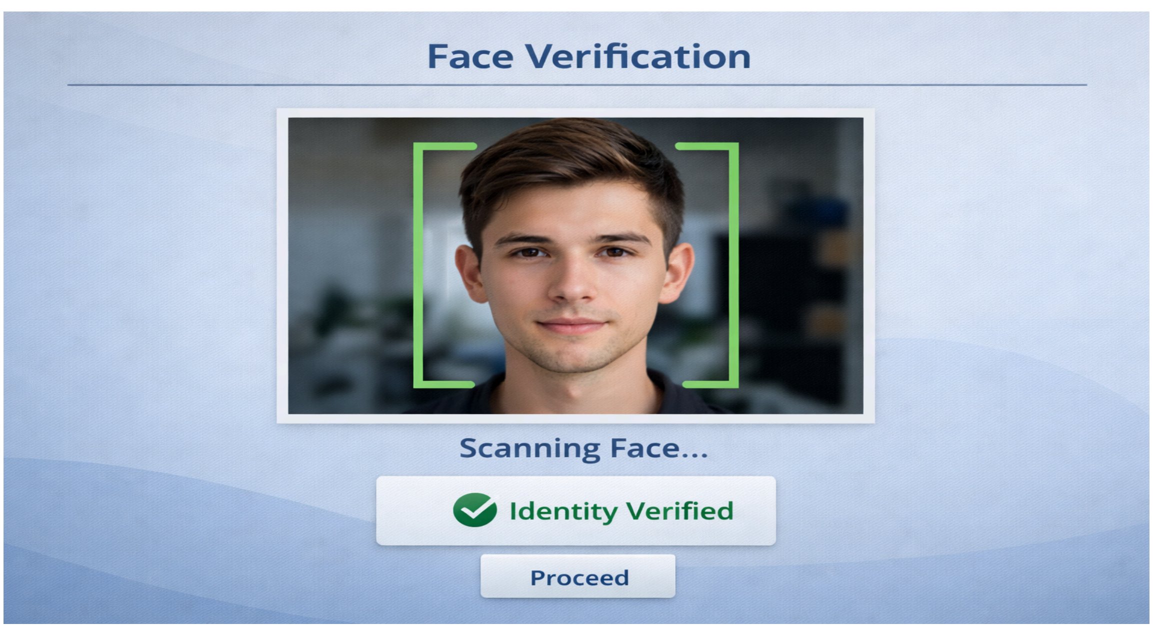
8. **List the Technical features and Elements of the invention along with the Description of your invention from start to end.**
The invention consists of multiple modules working together in an integrated architecture.
The HMIV-Net module performs multi-modal identity verification using deep learning.
The ARUPLA module learns and analyzes user behavior patterns over time.
The HAD-RSM module detects fraud using hybrid anomaly detection techniques.
The DTBRAA module calculates trust scores and makes final decisions.
The SDTEP module ensures secure storage and encryption of transaction data.
All modules are interconnected to provide a seamless and intelligent ration distribution system.

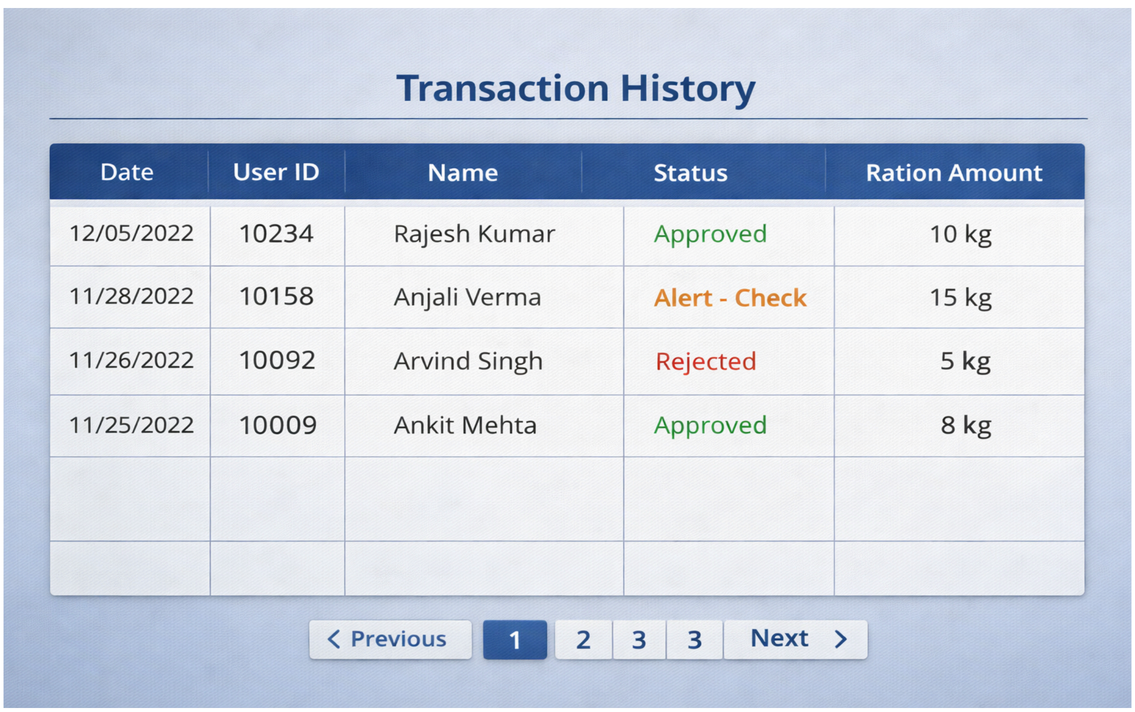
9. **List out the features of your invention which are believed to be new and distinguish them over the closest technology.**
The invention introduces a hybrid deep learning-based system combining multiple AI techniques.
It uniquely integrates identity verification with behavioral analysis and fraud detection.
The trust-based decision engine provides adaptive and dynamic decision-making.

The system continuously learns from user behavior to improve accuracy over time.
This combination of features is not present in existing ration distribution systems.

10. **Has the invention been built or tested or implemented? If so, please provide the particulars of the first time it was successfully built or implemented (when, where, by whom, and evidence of this event including written or on-line pointers to documentary evidence):All testing details**

The invention has been implemented using Python and deep learning frameworks such as TensorFlow or Keras.
The system has been tested using sample datasets for face recognition and transaction records.
Identity verification accuracy and fraud detection performance have been evaluated.
The results demonstrate improved efficiency and reduced fraud compared to traditional systems.




Transaction History

Date	User ID	Name	Status	Ration Amount
12/05/2022	10234	Rajesh Kumar	Approved	10 kg
11/28/2022	10158	Anjali Verma	Alert - Check	15 kg
11/26/2022	10092	Arvind Singh	Rejected	5 kg
11/25/2022	10009	Ankit Mehta	Approved	8 kg

11. Briefly state when and how you first conceived this idea?

The idea was conceived as part of an AI-assisted coding project.
It was developed to address real-world challenges in public distribution systems.
The concept evolved by integrating deep learning techniques with practical system design.

12. **Have you sold, offered for sale, publicly used or published anything related to this invention? If yes, please briefly explain the dates and circumstances. List those individuals to whom you have revealed your invention. Were non-discloser documents signed prior to discloser in each case? Please state any deadlines of which you may be aware for filing an application on this invention.**

    The invention has not been publicly disclosed or commercialized.
    No prior publications or sales related to this invention have been made.
    Confidentiality has been maintained throughout the development process.

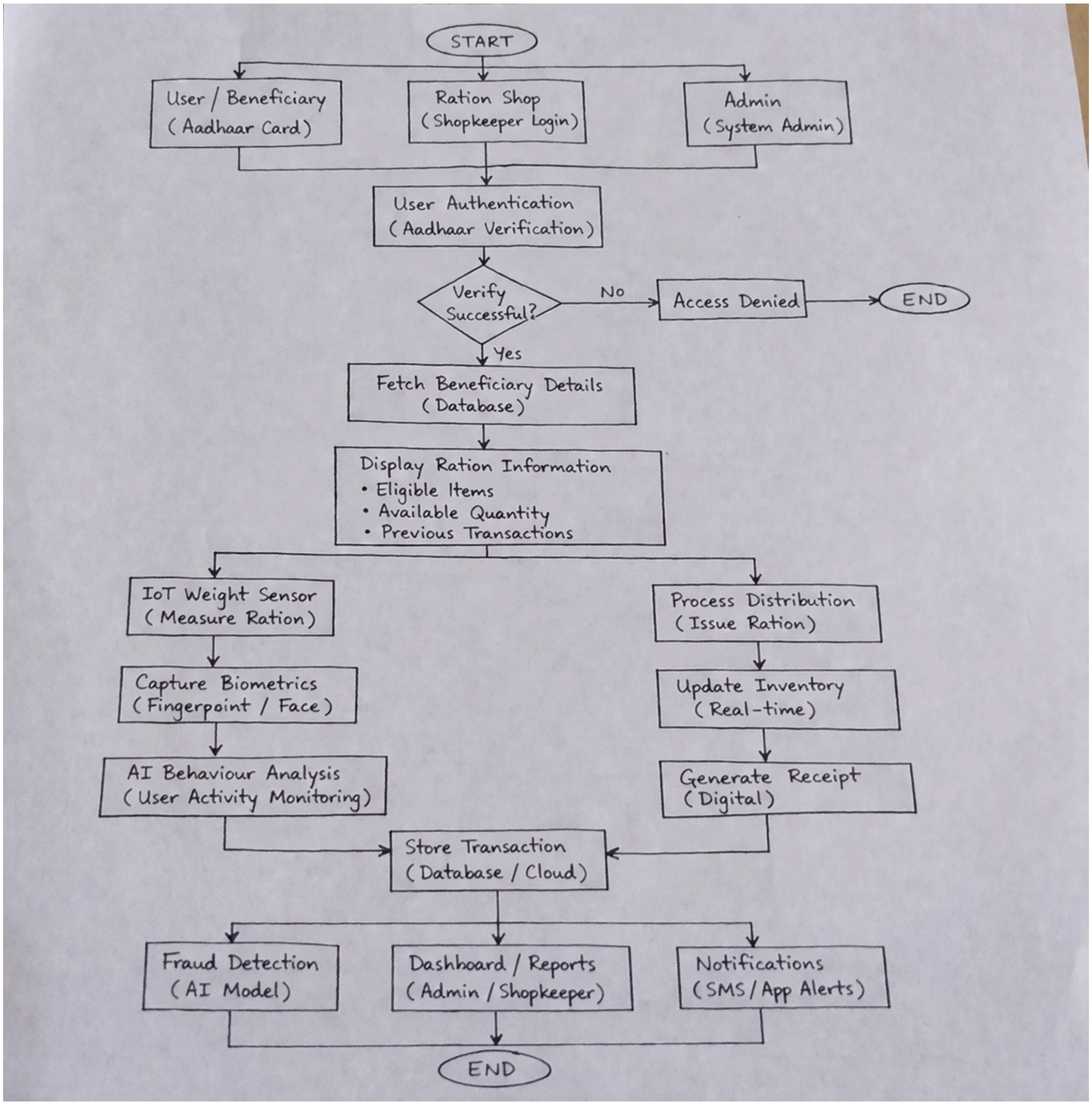
13. **Include any reasons that your invention would not have been obvious to someone of average skill in the art.**

    The invention is non-obvious as it combines multiple advanced technologies into a single system.The integration of identity verification, behavior analysis, and anomaly detection is unique.The trust-based decision-making approach introduces a novel method of resource allocation.Such a hybrid system is not evident from existing technologies.

14. **Additional comments by inventor (if you want to give more details out of scope of this IDF).**

    The system is scalable and can be deployed at a national level. Future enhancements may include blockchain integration for improved security.The invention can be extended to other domains such as healthcare and financial systems.

15. **Drawings/Flowchart/Table**



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